

Caracara Home Range Analysis

1. Objectives

The goal of this project was to assess the home ranges of four Caracaras in Florida as well as determine the Caracara's home ranges pre and post fledge. Once home ranges had been assessed this project quantified the composition of different vegetation types within each of the Caracara's home ranges using the 2011 National Land Cover Dataset (NLCD).

2. Data

The primary datasets used in this study were movement tracks from two female Caracaras and two male Caracaras. The data of this study consists of two breeding pairs, pair CC004 of female 36312 and male 36313 and of pair CC001 of female 44868 and male 68230. Breeding pair CC004 successfully fledged one chick on 05/02/11 while breeding pair CC001 successfully fledged one chick on 07/12/11. Female 36312 was first captured on 04/19/11 and data was recorded until unit failure on 05/22/11 resulting in 221 data points. Male 36313 was first captured on 04/15/11 and data was recorded until unit failure on 10/10/11 resulting in 1420 data points. Female 44868 was first captured on 06/07/11 and data was recorded until unit failure on 07/19/11 resulting in 40 data points. Male 68230 was first captured on 05/25/11 and data was recorded until unit failure on 10/09/11 resulting in 1098 data points. A secondary data set used in this was a NLCD raster from 2011 that covers the study area.

3. Data Preparation

The data was delivered in 4 .xls files, one for each bird. Each one of these files had two "large columns" of data that each consisted of multiple columns (date, time, lat, lon, ect...) with some summary statistics below these two "large columns". As far as we could tell these two "large columns" were identical and we copied one of the "large columns" worth of data into another .xls file that was then converted into a .csv to then be brought into R.

The NLCD raster was downloaded from the USGS earth explorer data portal for 2011 and from the tile covering the study area. This raster was brought into QGIS where it was then reprojected into UTM North Zone 17 and clipped to an AOI shapefile that roughly covered the study area.

4. Data Cleaning

All data cleaning was performed in the R script *prepare_transmitter_data_for_AKDE_home_range_analysis*. The following cleaning steps were taken and the following plots were created in this R script.

- Create a datetime variable for each Caracara.
- Every single 9pm time stamp for each Caracara appeared to have the wrong date. As a result, all rows with a datetime at 9pm were filtered out of the data set.
- The first days worth of data from each Caracara were removed so that the birds could rehabilitate after initial trapping.
- Histograms of the time gaps between points were plotted and can be seen in the "supplemental figures" section of this report.

- The median fix rate was calculated for each Caracara to confirm the 2 hour recording interval of the transmitters. The median fix rate value for each Caracara was 2 hours.
- Any rows with missing values in either the latitude or longitude column were filtered out.
- utm_x and utm_y columns were created for each Caracara by converting the WGS84 coordinate system into UTM North Zone 17.
- Step lengths were calculated from the UTM points.
- Histograms of the step lengths were plotted and can be seen in the “supplemental figures” section of this report. It did not seem unrealistic that a Caracara can travel multiple kilometers in 2 hours and as a result no filtering was done based on step length.
- The Caracara M68230 was the only individual that appeared to have a positional outlier and this single data point was filtered out.
- Plots for the Caracara’s utm_x and utm_y were plotted and can be seen in the “supplemental figures” section of this report.
- All datetime columns were filtered to remove any rows with a NA present in the datetime column.
- The utm_y was plotted over time and can be seen in the “supplemental figures” section of this report.
- The data was resampled into 2 hour intervals.
- A final filtering step was applied to remove any empty rows.
- A .csv for each Caracara was saved in “data/clean_data/” which were the .csvs used to assess the home ranges.
- .csvs were also made for each Caracara that were data from pre and post fledge. The pre-fledge data was saved in “data/clean_data/pre_fledge/” and were called “pre_bird-code”. The post-fledge data was saved in “data/clean_data/post_fledge/” and were called “post_bird-code”.
- The length of each final cleaned data set can be seen below.
 - The Caracara F44868 had a post-fledge data set of only 2 points and as a result F44868 did not have a post-fledge data set for further analysis.

Data Length	F36312	M36313	F44868	M68230
Entire Season	186	187	28	795
Pre-Fledge	68	112	26	169
Post-Fledge	118	75	2	626

5. AKDE (Autocorrelated kernel density estimator) Home Range Methods

The cleaned .csvs were brought into the *AKDE_home_range_analysis_save_ESRI_shp* R script where the following steps were taken.

- The variograms were assessed for each Caracara to ensure that individuals displayed home range movements (variograms plateau).

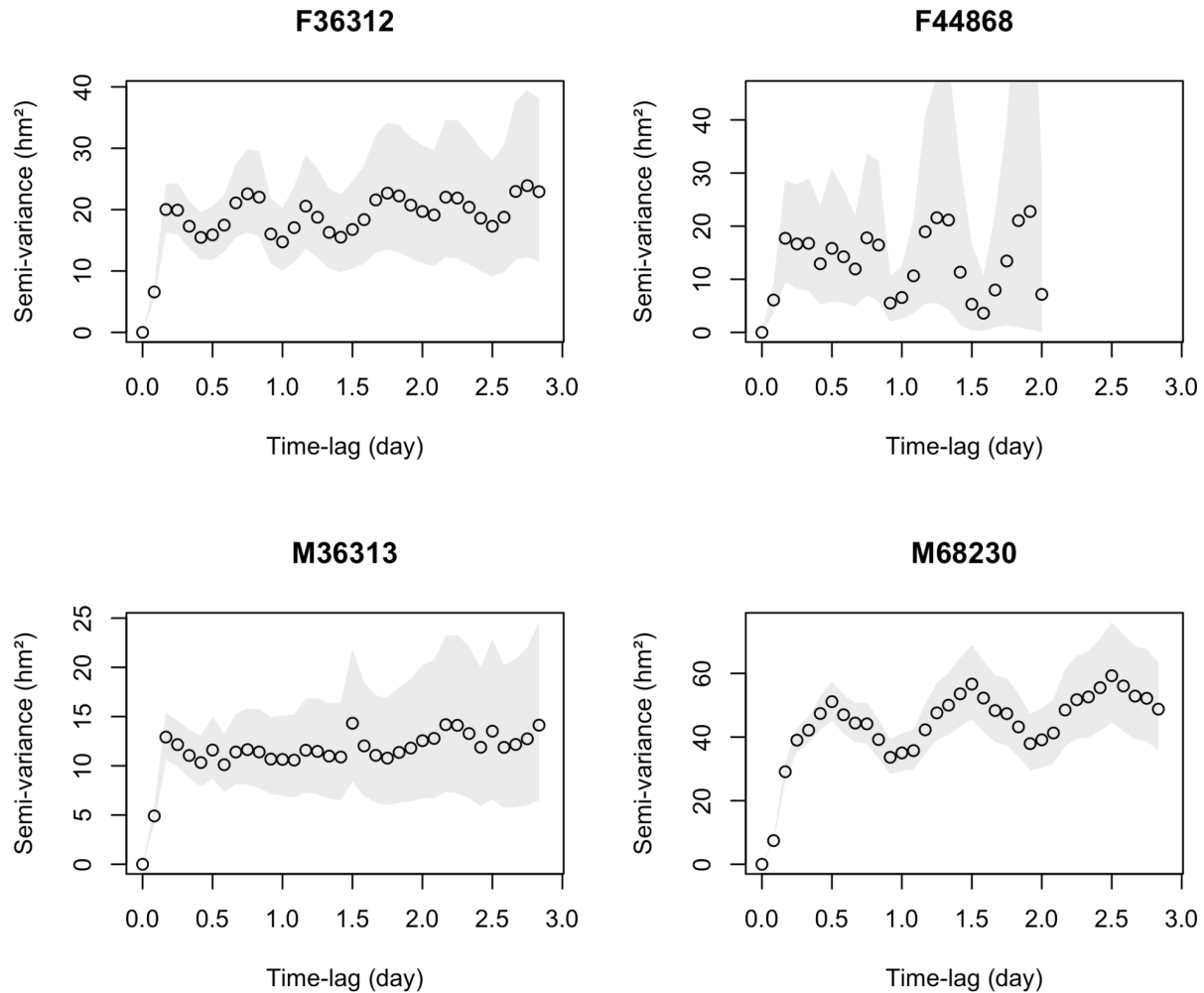


Fig. 1. Empirical variograms showing autocorrelation of data points collected for four individuals.

- To determine a suitable movement model for each Caracara, we compared movement models (OU: Ornstein-Uhlenbeck motion, OUF: OU motion including foraging, IID: uncorrelated KDE) and fitted as anisotropic (non-circular) or isotropic (circular) versions, which were compared with Akaike Information Criterion (AIC) using the ctmm package (Chris H. Fleming et al. 2014; C. H. Fleming et al. 2015).
- The best model (with lowest AIC) for each Caracara was plotted against the empirical variograms
 - OU anisotropic was the best model for $\frac{3}{4}$ Caracaras with only one having IID anisotropic as their best model. This bird however was quite data poor, only having 28 data points, and it was assumed the simple model was selected due to

sample size. We fit AKDE with an OU anisotropic model, which assumes that the animals return to a center of activity, to estimate home ranges for all 4 caracaras.

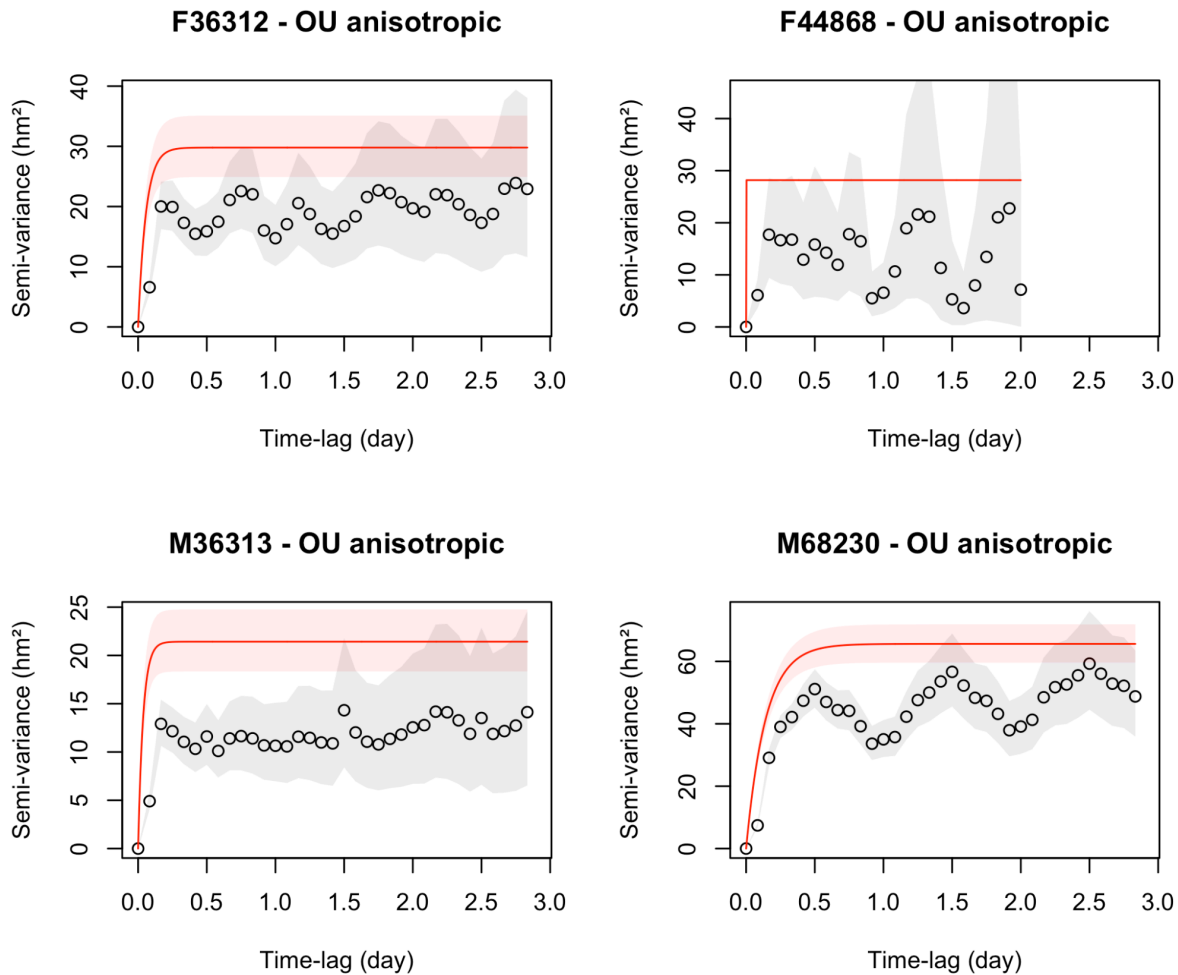


Fig. 2. Empirical variograms for four individuals with expected correlation from top model plotted as the red line. Note that only lags up to three days are reported so that the inflection from the curve can be easily observed. This figure should probably be included in the appendix

- An amt movement track was created for each Caracara
- The ou anisotropic model was fit to all individuals in amt
- The 95% and 50% isopleth home ranges were created for all individuals
- The Area for all home ranges was calculated in km²
- The home ranges were saved as ESRI shapefiles
 - Seasonal home ranges were saved to "/>
- The process of creating home ranges and determining home range statistics were repeated for the pre and post fledge movement tracks. Variograms and model assessment were not

repeated for pre and post fledge movement tracks. Instead the data was loaded in, converted from dataframes to movement tracks, and then the home ranges were created.

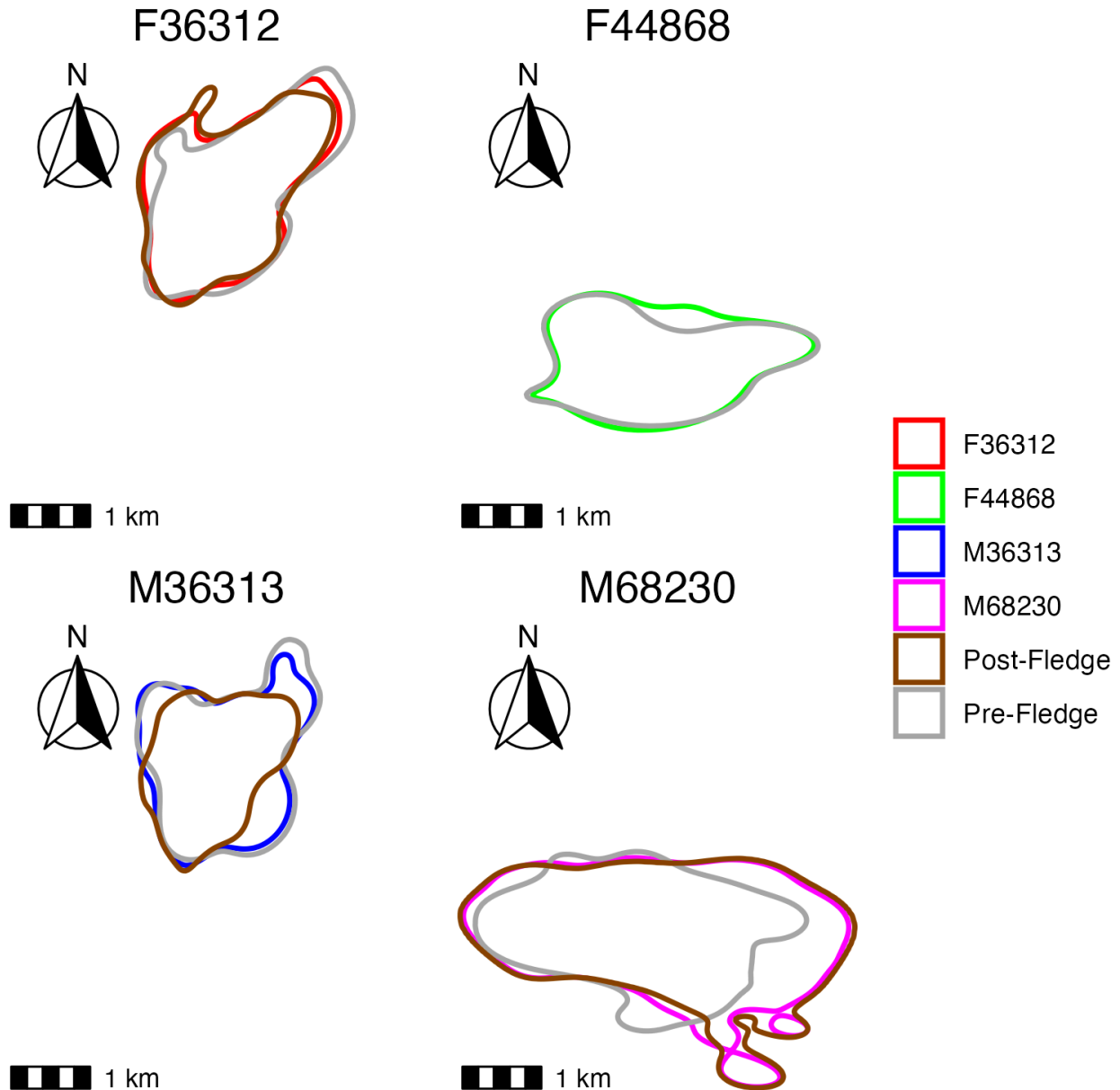
- Pre-fledge home ranges were saved to” /data/shp_files/pre_fledge/”
- Post-fledge home ranges were saved to” /data/shp_files/post_fledge/”
- Statistics for home range area can be seen in the Caracara_stats.xlsx as well as re-calculated in the *AKDE_home_range_analysis_save_ESRI_shp* R script

6. AKDE Home Range Results

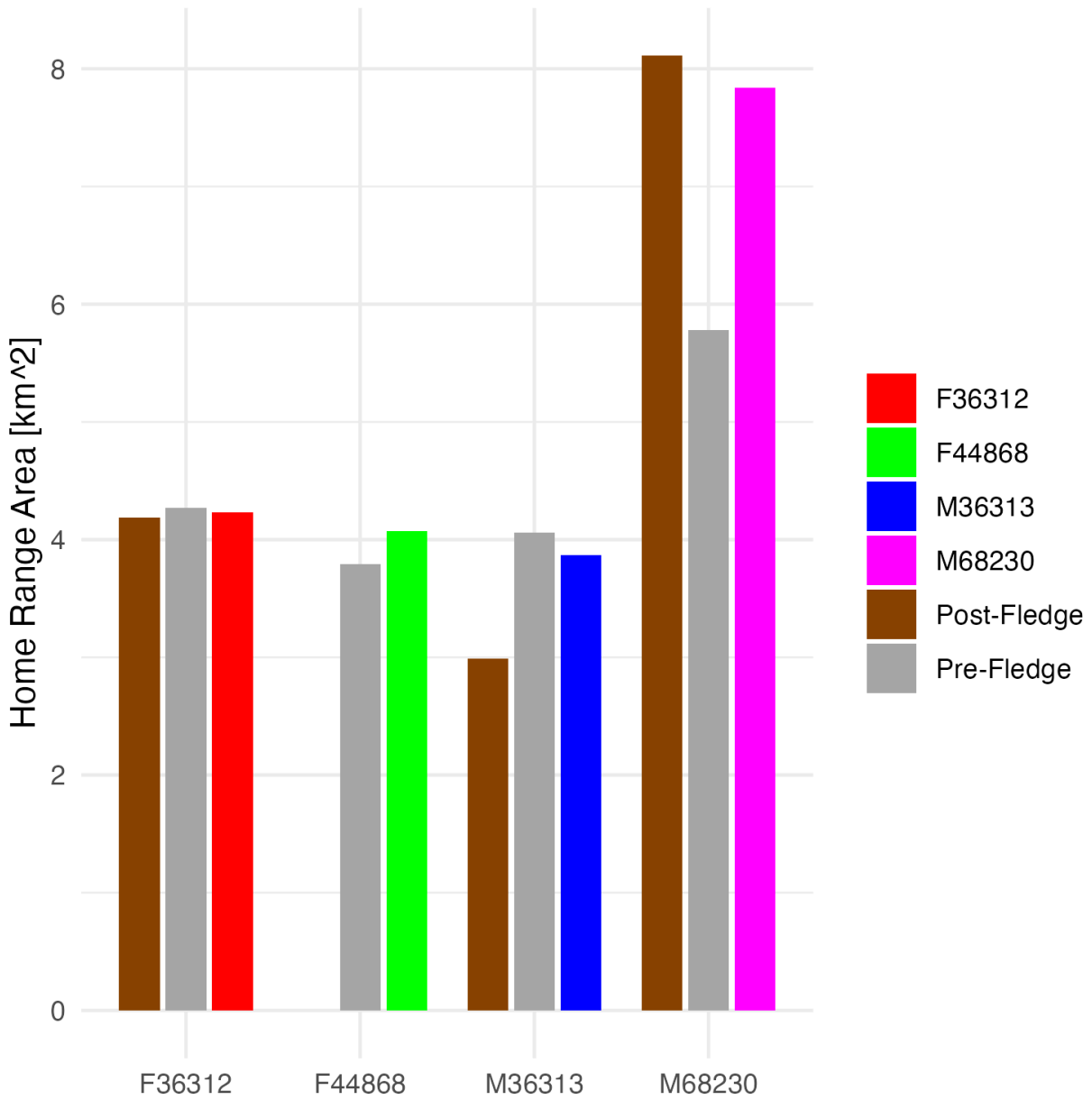
- All four Caracara seasonal home ranges (95% isopleths from AKDE fitted with OU anisotropic model) plotted together. These plots were created in the *all_on_one_home_range_plot* R script requiring the ESRI shape files for seasonal home ranges.



- The seasonal home range of each Caracara was plotted against its home range for pre and post fledge. The seasonal home ranges for each Caracara are colored the same as the above plot where each Caracara has a unique color. All pre-fledge home ranges are plotted gray and all post-fledge home ranges are plotted in brown. This plot was created in the *all_on_one_home_range_plot* R script requiring seasonal, pre-fledge, and post-fledge ESRI shape files as inputs.

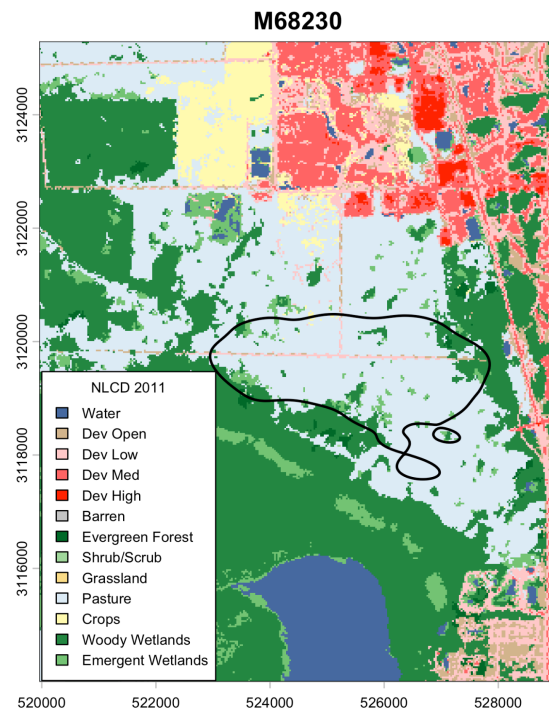
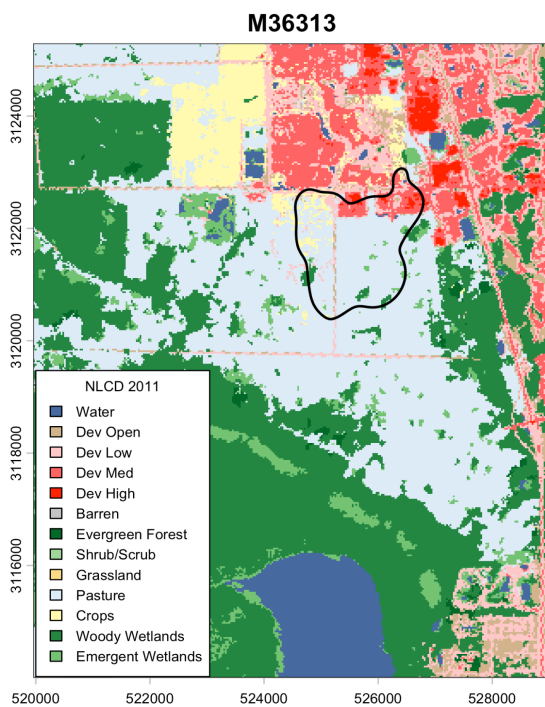
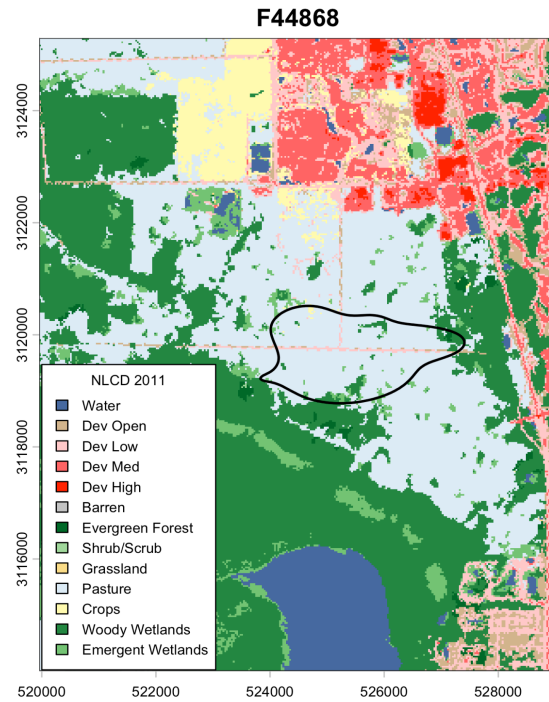
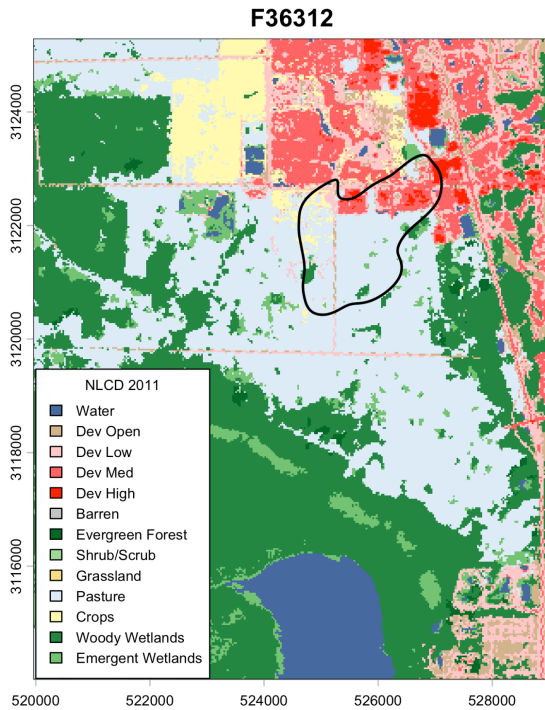


- The area for every home range shapefile can be visualized in the plot below. These bar charts have the same color pattern as defined above. This plot was created in the *home_range_area_compare_plot* R script. Area values were manually entered from Caracara_stats.xlsx.



7. Home Range Composition Results

- Each seasonal home range was plotted in black against the 2011 NLCD layer in the *analyze_home_range_shp_with_NLCD* R script. Inputs needed for this script are the seasonal home range shapefiles and the clipped NLCD raster.



- The percentage cover of each home range was calculated from the NLCD raster in the *analyze_home_range_shp_with_NLCD* R script. Pie charts were plotted for each individual which can be seen in the “supplemental figures” and the numerical results for this section can be found in *Caracara_stats.xlsx*. An example summary table for percentage cover can be seen below.

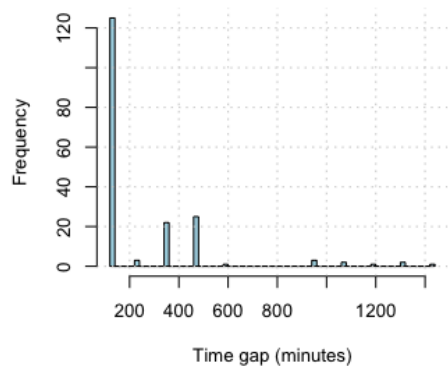
	F36312	F44868	M36313	M68230
Water	0.36	0	0.3	0
Dev Open	2.62	0.66	1.77	0.79
Dev Low	9.69	3.28	7.24	2.16
Dev Med	12.89	0	9.69	0
Dev High	2.43	0	1.47	0
Barren	0	0	0	0
Evergreen Forest	0	0.44	0	0.44
Shrub/Scrub	0	0	0	0.05
Grassland	0	0	0	0
Pasture	63.74	85.22	69.66	79.45
Crops	4.3	0.42	5.5	0.17
Woody Wetlands	2.39	6.71	2.45	12.06
Emergent Wetlands	1.58	3.26	1.93	4.89

8. References

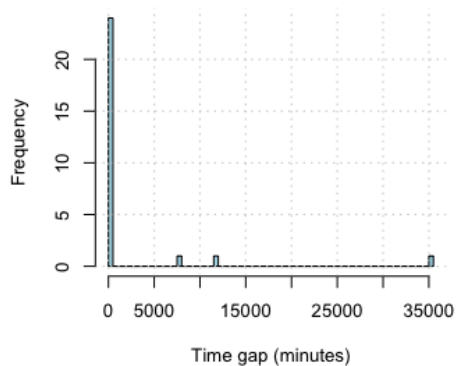
- Fleming, C. H., W. F. Fagan, T. Mueller, K. A. Olson, P. Leimgruber, and J. M. Calabrese. 2015. "Rigorous Home Range Estimation with Movement Data: A New Autocorrelated Kernel Density Estimator." *Ecology* 96 (5): 1182–88. <https://doi.org/10.1890/14-2010.1>.
- Fleming, Chris H., Justin M. Calabrese, Thomas Mueller, Kirk A. Olson, Peter Leimgruber, and William F. Fagan. 2014. "From Fine-Scale Foraging to Home Ranges: A Semivariance Approach to Identifying Movement Modes across Spatiotemporal Scales." *The American Naturalist* 183 (5): E154-167. <https://doi.org/10.1086/675504>.

9. Supplemental Figures

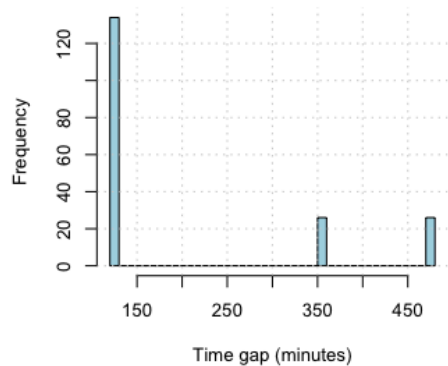
Time gaps (mins) for F36312



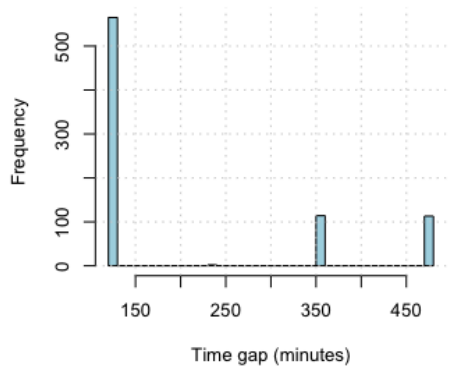
Time gaps (mins) for F44868



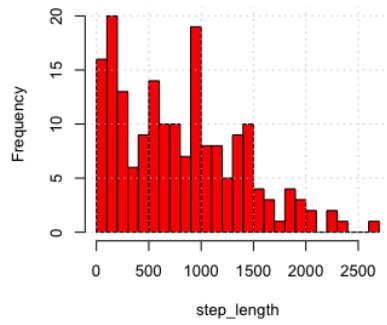
Time gaps (mins) for M36313



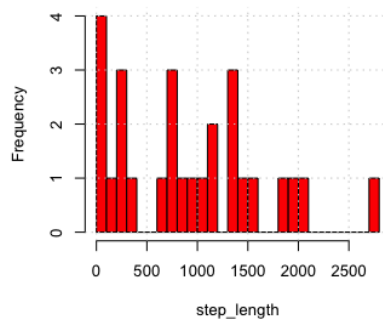
Time gaps (mins) for M68230



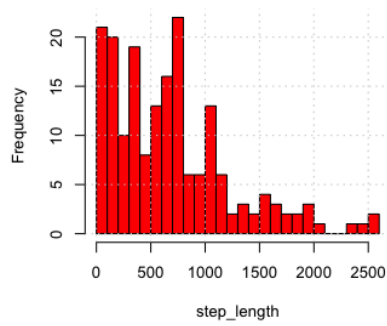
Step Length (meters) F36312



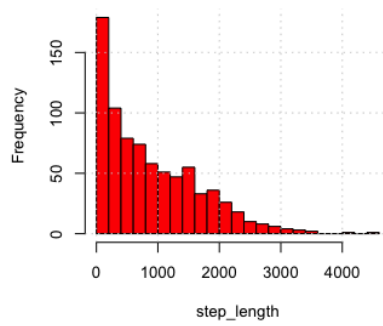
Step Length (meters) F44868



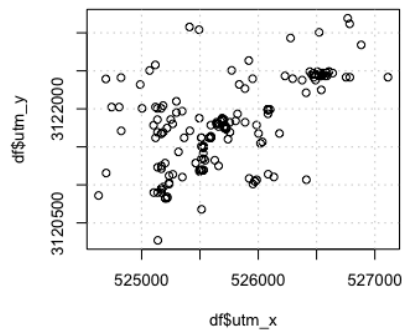
Step Length (meters) M36313



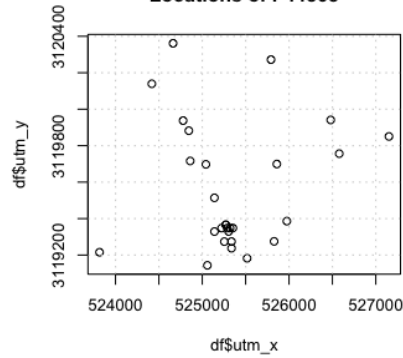
Step Length (meters) M68230



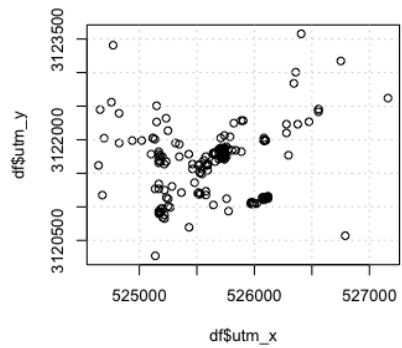
Locations of F36312



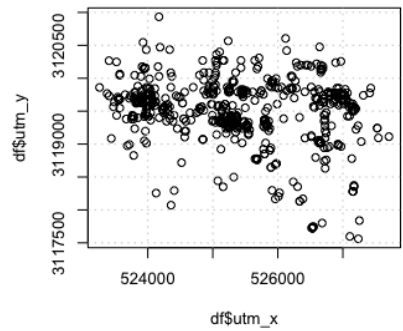
Locations of F44868



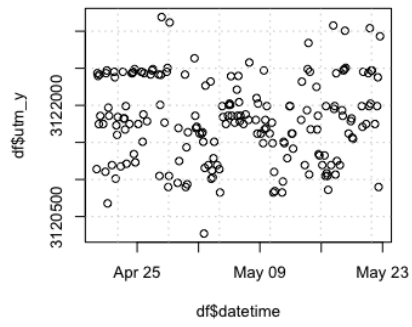
Locations of M36313



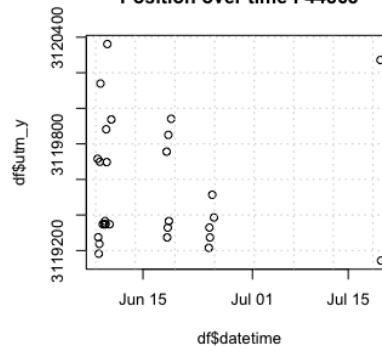
Locations of M68230



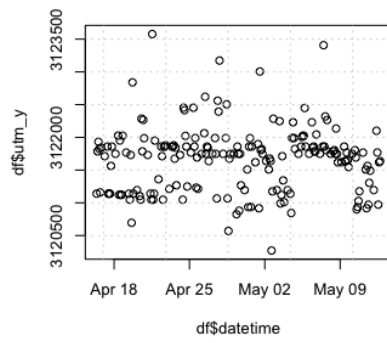
Position over time F36312



Position over time F44868



Position over time M36313



Position over time M68230

